Axel S. Martin

PhD Student · Biostatistics

New York University - Langone Health

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Education

New York University - Langone Health *New York City - NY*

PhD - Biostatistics *2021 - present*

- Advisors: Dr. Michele Santacatterina & Dr. Iván Díaz
- Topics: Machine learning causal inference methodology for doubly robust estimators

University of Michigan *Ann Arbor - MI*

MS - Biostatistics *2015 - 2017*

- Advisor: Shawn Lee
- Topics: Genome-Wide Association Study with a focus on height and Body Mass Index

McGill University *Montreal - QC*

BS - Mathematics *2011 - 2014*

- Major in Mathematics
- Minor in Computer Science

Professional Experience

Memorial Sloan Kettering Cancer Center *New York City - NY*

Research Biostatistician *2017 - 2022*

- Survival prediction and risk stratification Ensemble Machine Learning algorithms for cancer
- Genomic data processing and analysis software development
- Web-based application development, design and maintenance for statistical analysis
- GENIE multi-center genomic study, regimen specific progression free survival analysis
- Optimized sequential treatment regimes for advanced cancer patients based on actionable genomic targets

University of Michigan *Ann Arbor - MI*

Research Assistant *Summer 2016*

- Summer research project in Dr. Jeffrey Kidd's lab
- Building a Python/UNIX pipeline to download and process dog genomes data into analysis ready format
- Work on mapping, imputation problems and haplogroups creation
- Analysis of the canid Y-chromosome phylogeny

MiRCore *Ann Arbor - MI*

Data Analyst *2015-2016*

- Work-research program at MiRCore
- MicroRNA and RNA data analysis in order to study the link between AXIN2 gene and 3 microRNAs in colon cancer patients
- Use of principle component analysis and correlation heatmaps
- Web-based applications development

Publications

Axel Martin, Michele Santacatterina, Iván Díaz. Non-parametric efficient estimation of marginal structural models with continuous time-varying treatments, Biometrika, 2026; doi

Ronglai Shen, **Axel Martin** and al. Harnessing Clinical Sequencing Data for Survival Stratification of Patients with Metastatic Lung Adenocarcinomas. JCO Precision Oncology 2019 :3, 1-9.

Matthew T Oetjens, **Axel Martin**, Krishna R Veeramah, Jeffrey M Kidd. Analysis of the canid Y-chromosome phylogeny using short-read sequencing data reveals the presence of distinct haplogroups among Neolithic European dogs. BMC Genomics. 2018 May 10;19(1):350. doi: 10.1186/s12864-018-4749-z.

Marjorie G Zauderer, Axel Martin and al. The use of a next-generation sequencing-derived machine-learning risk-prediction model (OncoCast-MPM) for malignant pleural mesothelioma: a retrospective study. The Lancet Digital Health. July 28, 2021: Volume 3, DOI: 10.1016/S2589-7500(21)00104-7

Hyungrok Do, Preston Putzel, **Axel Martin**, Padhraic Smyth, Judy Zhong. Fair Generalized Linear Models with a Convex Penalty. Proceedings of the 39th International Conference on Machine Learning, PMLR 162:5286-5308, 2022.

Sam Whipple, **Axel Martin** and al. Validation of broad panel clinical sequencing-based genomic risk stratification in patients with advanced lung adenocarcinomas. DOI: 10.1200/JCO.2019.37.15-suppl.9113 Journal of Clinical Oncology 37, no. 15-suppl (May 20, 2019) 9113-9113.

DD Correa, J Satagopan, **A Martin**, E Braun, and al. Genetic variants and cognitive functions in patients with brain tumors. Neuro-oncology 21 (10), 1297-1309

Saptarshi Chakraborty, **Axel Martin**, Zoe Guan, Colin B Begg, Ronglai Shen. Mining mutation contexts across the cancer genome to map tumor site of origin. Nature Communications 12, Article number: 3051 (2021)

Ori Barzilai, **Axel Martin**, Anne S Reiner, Ilya Laufer, Adam Schmitt, Mark H Bilsky. Clinical reliability of genomic data obtained from spinal metastatic tumor samples. Neuro-Oncology, Volume 24, Issue 7, July 2022, Pages 1090–1100, https://doi.org/10.1093/neuonc/noac009

Sean M Devlin, **Axel Martin**, Irina Ostrovnaya. Identifying prognostic pairwise relationships among bacterial species in microbiome studies. November 9, 2021. DOI: https://doi.org/10.1371/journal.pcbi.1009501

Presentations

** presenting author; ⁺ mentored undergraduate*

Contributed Presentations

2024. *Non-parametric efficient estimation of marginal structural models with multi-valued time-varying treatments*. JSM

2024. *Non-parametric efficient estimation of marginal structural models with multi-valued time-varying treatments*. ACIC

2023. *Efficient generalizability and transportability of survival causal effects*. JSM

2019. *how to use clinical sequencing data for survival stratification with metastatic lung cancer*. JSM.

2019. *OncoCast: an R package which is a flexible ensemble machine learning framework for survival prediction.*. IMSCO.

Teaching Experience

Fall 2023	**Causal inference I**, Teaching Assistant
Fall 2016	**Introduction to statistics**, Teaching Assistant